

JSS 3 MATHEMATICS – FIRST TERM

TOPIC 1: WHOLE NUMBERS I

1. Binary Number System

The binary system uses only two digits: 0 and 1. It is the number system used by computers. Each position is a power of 2. Example: $1011_{\text{two}} = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 11_{\text{ten}}$. To convert from base 10 to binary, divide by 2 repeatedly and read remainders from bottom to top.

2. Using Computers for Simple Calculations

Calculators and computers perform arithmetic quickly and accurately. Always check that you have entered the correct numbers and operations. Understand the order of operations (BODMAS/BIDMAS) when interpreting results.

3. Translation of Word Problems into Numerical Expressions

Read the problem carefully; identify what is given and what is required; assign symbols if needed; write a mathematical expression or equation; solve and check that the answer makes sense in the context of the problem.

4. Example

"Five more than twice a number is 17." $\rightarrow 2x + 5 = 17 \rightarrow 2x = 12 \rightarrow x = 6$.

5. Summary

Binary uses 0 and 1; computers rely on it. Translating word problems into expressions is a key skill for solving real-life maths questions.

End of Topic 1